Retarded fields from potentials

$$\vec{E} = -\vec{\nabla}\phi - \frac{\partial \vec{A}}{\partial t}$$

$$\vec{B} = \vec{\nabla}\times\vec{A}$$

$$\varphi(\vec{r},t) = \frac{1}{4\pi\varepsilon_0}\int \frac{\rho(\vec{r}\,',t-\frac{R}{c})}{R}\,d\vec{r}\,' \qquad \vec{R} = \vec{r}-\vec{r}\,'$$

$$\vec{A}(\vec{r},t) = \frac{\mu_0}{4\pi}\int \frac{\vec{j}(\vec{r}\,',t-\frac{R}{c})}{R}\,d\vec{r}\,'$$

$$\vec{\nabla}_r\varphi = \frac{1}{4\pi\varepsilon_0}\int\left(\left(\vec{\nabla}_r\rho\right)\frac{1}{R} + \rho\,\vec{\nabla}_r\frac{1}{R}\right)d\vec{r}\,'$$

$$\vec{\nabla}\frac{1}{R} = -\frac{1}{R^2}\vec{\nabla}R = -\frac{1}{R^2}\frac{\vec{R}}{R} = -\frac{\vec{R}}{R^3}$$

$$\vec{\nabla}_r\rho(\vec{r}\,',t-\tfrac{R}{c}) = \frac{\partial\rho}{\partial t}\vec{\nabla}_r\left(t-\frac{R}{c}\right) = \frac{\partial\rho}{\partial t}\left(-\frac{\vec{R}}{Rc}\right)$$

$$\vec{\nabla}\varphi = \frac{-1}{4\pi\varepsilon_0}\int\left(\frac{\rho(\vec{r}\,',t-\frac{R}{c})\,\vec{R}}{R^3} + \frac{\partial\rho(\vec{r}\,',t-\frac{R}{c})}{\partial t}\frac{\vec{R}}{R^2c}\right)d\vec{r}\,'$$

$$\vec{\nabla}\times\vec{A} = \frac{\mu_0}{4\pi}\int\left(\vec{\nabla}\left(\frac{1}{R}\right)\times\vec{j} + \frac{\vec{\nabla}\times\vec{j}}{R}\right)d\vec{r}\,'$$

$$\vec{\nabla}\times\vec{j}(\vec{r}\,',t-\tfrac{R}{c}) = \hat{e}_a\,\varepsilon_{abc}\frac{\partial}{\partial r_b}j_c(\vec{r}\,',t-\tfrac{R}{c})$$

$$= \hat{e}_a\,\varepsilon_{abc}\frac{\partial j_c}{\partial t}\left(-\frac{1}{c}\frac{\partial R}{\partial r_b}\right) = \hat{e}_a\,\varepsilon_{abc}\frac{R_b}{R}\frac{\partial j_c}{\partial t}\left(-\frac{1}{c}\right)$$

$$\frac{\partial R}{\partial r_b} = \frac{\partial}{\partial r_b}\sqrt{\sum_a R_aR_a} = \frac{1}{2\sqrt{\sum_a R_aR_a}}\frac{\partial}{\partial r_b}\sum_a R_aR_a = \frac{R_b}{R}$$

$$\vec{\nabla}\times\vec{A} = -\frac{\mu_0}{4\pi}\int\left(\frac{\vec{R}\times\vec{j}}{R^3} + \frac{\vec{R}\times\frac{\partial\vec{j}}{\partial t}}{cR^2}\right)d\vec{r}\,' \quad \text{where } \vec{j}(\vec{r}\,',t-\tfrac{R}{c}).$$

Continuity eqn $\frac{\partial \rho}{\partial t} + \vec{\nabla}\cdot\vec{J} = 0$

but be careful because of retardation:

$$\underset{\frac{\partial}{\partial \vec{r}\,'}}{\vec{\nabla}'}\cdot\vec{J}\left(\vec{r}, t-\frac{R}{c}\right) = \underbrace{\frac{\partial \vec{J}}{\partial \vec{r}\,'}}_{-\frac{\partial \rho}{\partial t}} + \frac{\partial \vec{J}}{\partial t}\cdot\left(-\frac{1}{c}\underbrace{\vec{\nabla}' R}_{-\frac{\vec{R}}{R}}\right)$$

describes retardation

describes flux through boundary

$\hat{R} = \frac{\vec{R}}{R}$

Hence $\int \frac{\partial \rho}{\partial t}\frac{\hat{R}}{Rc}\,d\vec{r}\,' = -\frac{1}{c}\int d\vec{r}\,'\frac{(\vec{\nabla}'\cdot\vec{J})\hat{R}}{R} + \frac{1}{c^2}\int d\vec{r}\,'\frac{\partial \vec{J}\cdot\hat{R}}{\partial t}\frac{\hat{R}}{R}$

Consider $\frac{\vec{\nabla}'\cdot\vec{J}}{R}\hat{R} = \frac{\partial \vec{J}_j}{\partial r_j'}\frac{R_i}{R^2} = \frac{\partial}{\partial r_j'}\left(\frac{J_j R_i}{R^2}\right) -$

$$- J_j\frac{\partial}{\partial r_j'}\left(\frac{R_i}{R^2}\right) = \frac{\partial}{\partial \vec{r}_j\,'}\left(\frac{J_j R_i}{R^2}\right) + \frac{J_i - 2\vec{J}\cdot\hat{R}\,\hat{R}_i}{R^2}$$

$$\frac{\partial}{\partial r_j'}\frac{R_i}{R^2} = \frac{-\delta_{ij}}{R^2} - \frac{2R_i}{R^3}\underbrace{\frac{\partial R}{\partial r_j'}}_{-\frac{R_j}{R}} = \frac{-\delta_{ij} + 2\hat{R}_i\hat{R}_j}{R^2}$$

$$\Rightarrow \int d\vec{r}\,'\frac{\partial}{\partial \vec{r}_j}\left(\frac{J_j R_i}{R^2}\right) = 0 \text{ (surface integral)}$$

$$-\frac{1}{c}\int d\vec{r}\,'\frac{\vec{\nabla}'\cdot\vec{J}}{R}\hat{R} = \frac{1}{c}\int d\vec{r}\,'\frac{2(\vec{J}\cdot\hat{R})\hat{R} - \vec{J}}{R^2} =$$

$A\times(B\times C) = B(A\cdot C) - C(A\cdot B)$

$$= \frac{1}{c}\int d\vec{r}\,'\frac{(\vec{J}\cdot\hat{R})\hat{R} + (\vec{J}\times\hat{R})\times\hat{R}}{R^2}$$

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\int d\vec{r}\,'\left\{\frac{\rho\,\hat{R}}{R^2} + \frac{(\vec{J}\cdot\hat{R})\hat{R} + (\vec{J}\times\hat{R})\times\hat{R}}{cR^2} + \frac{\left(\frac{\partial\vec{J}}{\partial t}\cdot\hat{R}\right)\hat{R}}{c^2R}\right\} - \frac{\mu_0}{4\pi}\int d\vec{r}\,'\,\frac{\frac{\partial\vec{J}}{\partial t}}{R}$$

$$\mu_0 = \frac{1}{\varepsilon_0 c^2}$$

But $\left(\frac{\partial\vec{J}}{\partial t}\times\hat{R}\right)\times\hat{R} = -\left(\hat{R}\times\frac{\partial\vec{J}}{\partial t}\right)\times\hat{R} =$

$$= (-)\left\{\frac{\partial\vec{J}}{\partial t} - \hat{R}\left(\hat{R}\cdot\frac{\partial\vec{J}}{\partial t}\right)\right\}$$

$$\Rightarrow \vec{E}(\vec{r},t) = \frac{1}{4\pi\varepsilon_0}\int d\vec{r}\,'\left\{\frac{\rho\,\hat{R}}{R^2} + \frac{(\vec{J}\cdot\hat{R})\hat{R} + (\vec{J}\times\hat{R})\times\hat{R}}{cR^2} + \frac{\left(\frac{\partial\vec{J}}{\partial t}\times\hat{R}\right)\times\hat{R}}{c^2R}\right\}$$

$$\vec{B}(\vec{r},t) = +\frac{\mu_0}{4\pi}\int d\vec{r}\,'\left\{\frac{\vec{J}\times\hat{R}}{R^2} + \frac{\frac{\partial\vec{J}}{\partial t}\times\hat{R}}{cR}\right\}$$

(These are Jefimenko equations)

Most important features: (1) Near fields $\sim \frac{1}{R^2}$

(2) Far fields $\sim \frac{1}{R}$ $\Rightarrow$ this is radiation.

(3) In the far region only <u>transverse</u> components of the current contribute to <u>radiation</u> fields $\vec{E}$ and $\vec{B}$.

<u>Far-fields</u> $\left(\sim \frac{1}{R}\right)$

$$\vec{E}(\vec{r},t) = \frac{-1}{4\pi\varepsilon_0}\int d\vec{r}\,' \; \frac{\frac{\partial}{\partial t}\left[\vec{J}(\vec{r}\,',t-\frac{R}{c}) - \left(\vec{J}(\vec{r}\,',t-\frac{R}{c})\cdot\hat{R}\right)\hat{R}\right]}{c^2 R}$$

$$= + \frac{\mu_0}{4\pi}\int d\vec{r}\,' \; \frac{\left(\frac{\partial}{\partial t}\vec{J}(\vec{r}\,',t-\frac{R}{c})\times\hat{R}\right)\times\hat{R}}{R}$$

$$\vec{B}(\vec{r},t) = \frac{\mu_0}{4\pi}\int d\vec{r}\,' \; \frac{\frac{\partial}{\partial t}\vec{J}(\vec{r}\,',t-\frac{R}{c})\times\hat{R}}{cR} =$$

$$= \frac{\mu_0}{4\pi c}\int d\vec{r}\,' \; \frac{\frac{\partial}{\partial t}\left[\vec{J}(\vec{r}\,',t-\frac{R}{c}) - \left(\vec{J}(\vec{r}\,',t-\frac{R}{c})\cdot\hat{R}\right)\hat{R}\right]\times\hat{R}}{R}$$

Both fields are expressed in terms of $\frac{\partial}{\partial t}\vec{J}_\perp(\vec{r}\,',t-\frac{R}{c})$

where $\vec{J}_\perp \equiv \vec{J} - (\vec{J}\cdot\hat{R})\hat{R} = -(\vec{J}\times\hat{R})\times\hat{R} = (\hat{R}\times\vec{J})\times\hat{R}$

is transverse current.

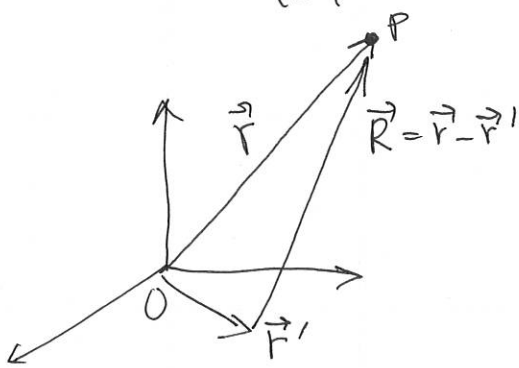
$\vec{J}_\perp$ is vector orthogonal to $\hat{R}$.

Hence we see: (a) $\vec{E}\perp\hat{R}$, $\hat{B}\perp\hat{R}$ and $\vec{E}\perp\vec{B}$

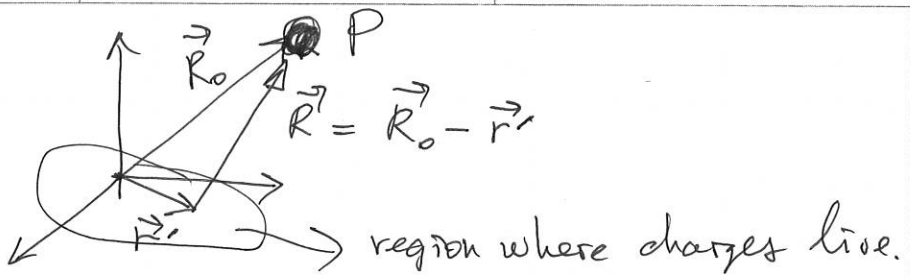
(plane wave structure)

(b) $\frac{|\vec{E}|}{|\vec{B}|} = c$. $\left(\text{for } \vec{R} = \vec{r}-\vec{r}\,' \approx \vec{r}\right)$

$r, R \gg r'$



let now



For $R_0 \gg r'$ (far field condition) $\vec{R} \approx \vec{R}_0$

$\Rightarrow \hat{R} = \hat{R}_0$ in which case (let $\hat{n} = \hat{R}_0 = \frac{\vec{R}_0}{R_0}$)

$$\vec{B} \cong \frac{\mu_0}{4\pi} \frac{1}{c} \int d\vec{r}' \, \frac{1}{R_0} \, \frac{\partial \vec{J}(\vec{r}', t - \frac{R}{c})}{\partial t} \times \hat{n} =$$

$$= \frac{1}{c} \frac{\partial}{\partial t} \vec{A}(\vec{r}, t) \times \hat{n}$$

$$\vec{E}(r,t) \approx \left\{ \left(\frac{1}{4\pi\varepsilon_0} \int d\vec{r}' \, \frac{\frac{\partial \vec{J}}{\partial t}}{c^2 R_0} \right) \times \hat{n} \right\} \times \hat{n}$$

$$\left(\frac{1}{\varepsilon_0 c^2} = \mu_0 \right) \quad = \left\{ \frac{\partial}{\partial t} \vec{A}(r,t) \times \hat{n} \right\} \times \hat{n} = c\vec{B} \times \hat{n}$$

so that $\vec{B} = \frac{1}{c} \hat{n} \times \vec{E}$.

Note that although $\frac{1}{R_0}$ dependence is factored out from under the integral, $\vec{E}(r,t)$ and $\vec{B}(r,t)$ still depend on $\vec{r} \equiv \vec{R}_0$ via the retarded time argument $t - \frac{R}{c} = t - \frac{|\vec{R}_0 - \vec{r}'|}{c} = t - \frac{|\vec{r} - \vec{r}'|}{c}$.

$$\vec{E} = c\,\vec{B}\times\hat{n}$$

$$\vec{B} = \frac{1}{c}\,\hat{n}\times\vec{E} = \frac{c}{c}\left(\hat{n}\times\vec{B}\times\hat{n}\right) = \vec{B}\,(\hat{n}\cdot\hat{n}) - \underbrace{(\vec{B}\cdot\hat{n})}_{0}\,\hat{n}$$

<u>Far-field</u>

$$R = |\vec{R}_0 - \vec{r}| = \left[(\vec{R}_0 - \vec{r})\cdot(\vec{R}_0 - \vec{r})\right]^{1/2} = \left(R_0^2 + r^2 - 2\vec{R}_0\cdot\vec{r}\right)^{1/2}$$

$$= R_0\left(1 - \frac{2\vec{R}_0\cdot\vec{r}}{R_0^2} + \frac{r^2}{R_0^2}\right)^{1/2} \approx R_0\left(1 - \frac{\vec{R}_0\cdot\vec{r}}{R_0^2}\right) = R_0 - \frac{\vec{R}_0\cdot\vec{r}}{R_0}$$

$$\Rightarrow \begin{cases} R \approx R_0 - \hat{n}\cdot\vec{r} \quad \left(\hat{n} = \dfrac{\vec{R}_0}{R_0}\right), \quad \dfrac{1}{R} \approx \dfrac{1}{R_0} \\ \hat{R} \approx \hat{n} \end{cases}$$

(corrections go $\frac{1}{R_0^2}$!)

$$\vec{B}(\vec{r},t) = \frac{\mu_0}{4\pi}\int d\vec{r}\,' \; \frac{\frac{\partial}{\partial t}\vec{J}(\vec{r}\,',t-\frac{R}{c})\times\hat{R}}{c\,R}$$

$$\approx \frac{\mu_0}{4\pi c R_0}\int d\vec{r}\,' \; \frac{\partial}{\partial t}\vec{J}\left(\vec{r}\,',t-\frac{R_0}{c}+\hat{n}\cdot\vec{r}\,'\right)\times\hat{n}$$

$$\vec{E}(\vec{r},t) \cong \frac{\mu_0}{4\pi R_0}\int d\vec{r}\,' \; \frac{\partial}{\partial t}\left\{\vec{J}\left(\vec{r}\,',t-\frac{R_0}{c}+\hat{n}\cdot\vec{r}\,'\right)\times\hat{n}\right\}\times\hat{n}$$

$$= c\,\vec{B}(\vec{r},t)\times\hat{n}.$$

But notice that $\vec{A}(\vec{r},t) \cong \frac{\mu_0}{4\pi}\int \frac{\vec{J}(\vec{r}\,',t-\frac{R_0}{c}+\hat{n}\cdot\vec{r}\,')}{R_0}$

under the same approximations

$$\Rightarrow \begin{cases} \vec{B} = \dfrac{1}{c}\,\dfrac{\partial\vec{A}(\vec{r},t)}{\partial t}\times\hat{n} \\ \vec{E} = c\,\vec{B}\times\hat{n} \end{cases}$$

need to know only $\vec{A}$.

Pointing vector $\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B} = \frac{c}{\mu_0} \vec{B} \times \hat{n} \times \vec{B}$

$$\vec{E} \times \frac{1}{c} \hat{n} \times \vec{E} = \frac{\hat{n}}{c}\left(\vec{E}^2\right) - \frac{\vec{E}}{c}\left(\hat{n} \cdot \vec{E}\right) = \frac{1}{c} \hat{n} E^2$$

$$\vec{S} = \frac{1}{\mu_0 c} E^2 \hat{n} = \frac{1}{\mu_0 c} c^2 B^2 \hat{n}$$

$$= c\, u\, \hat{n} \quad , u = \frac{1}{2}\left(\varepsilon_0 E^2 + \underbrace{\frac{1}{\mu_0} B^2}_{\frac{1}{\mu_0 c^2} E^2}\right) = \varepsilon_0 E^2$$

$$\Rightarrow \vec{S} = c\, u\, \hat{n}.$$

also $\vec{S} = \frac{c}{\mu_0} \vec{B} \times \hat{n} \times \vec{B} = \frac{c}{\mu_0} \vec{B}^2$

$$\frac{1}{\mu_0 c} = \cancel{\mu_0} \varepsilon_0 \frac{c}{\mu_0 c^2} = \varepsilon_0 c$$

Intensity into spherical angle $d\Omega$:

$$dI = \vec{S} \cdot d\vec{f} = S\, R_0^2\, d\Omega = c\, \varepsilon_0 E^2 R_0^2\, d\Omega$$

$$d\vec{f} = R_0^2\, d\Omega$$

Since $E, B \sim \frac{1}{R_0}$, $E^2 R_0^2 = \text{const}$

$\Rightarrow$ energy flows out with speed $\underline{c}$

$\Rightarrow$ radiation from accelerating charges.

Dipole radiation appears when we can neglect $\frac{\hat{n}\cdot\vec{r}}{c}$ in argument of $\rho, \vec{J}$ (via t') and $\vec{A}$.

let T be time when charge distribution changes

let S be size of the system.

$$\Rightarrow \frac{\vec{n}\cdot\vec{r}}{c} \sim \frac{S}{c} \ll T \Rightarrow S \ll cT \sim \frac{c}{\omega} \sim \lambda$$

wavelength

$\Rightarrow$ size $\underline{\underline{S \ll \lambda}}$

$\omega \sim \frac{1}{T}$ because it is acceleration that matters!

Also, $T \sim \frac{S}{v} \Rightarrow \lambda \sim \frac{cS}{v}$

$\Rightarrow$ from $S \ll \lambda$ follows $\underline{\underline{v \ll c}}$

Thus $R_0 \gg r$ and $R_0 \gg \lambda$ (neglect $\frac{\hat{n}\cdot\vec{r}}{c}$)

$$\Rightarrow \vec{A} = \frac{\mu_0}{4\pi R_0}\int d\vec{r}\, \vec{J}\left(\vec{r}, t - \frac{R_0}{c}\right) \qquad t' = t_0 - \frac{R_0}{c}$$

independent of $\vec{r}$!

$$\text{but } \vec{J} = \sum_j \vec{v}_j q_j \delta(\vec{r} - \vec{r}_j(t)) = \rho\vec{v}$$

$$\Rightarrow \vec{A} = \frac{\mu_0}{4\pi R_0}\left(\sum_j q_j \vec{v}_j\right)_{t'} = \frac{\mu_0}{4\pi}\frac{\dot{\vec{d}}}{R_0}$$

$$\text{but } \sum q_j \vec{v}_j = \frac{d}{dt}\sum_j q_j \vec{r}_j = \dot{\vec{d}} = \text{dipole moment}$$

$$\vec{B} = \frac{1}{c}\dot{\vec{A}} \times \hat{n} = \frac{\mu_0}{4\pi c R_0}\left(\ddot{\vec{d}} \times \hat{n}\right)$$

$$\vec{E} = c\vec{B} \times \hat{n}$$